PRODUCT TEARDOWN · PAYMENTS IN INDIA

Card checkout vs UPI

Why paying online with a Mastercard in India takes more steps than UPI, and how passkeys could close the gap

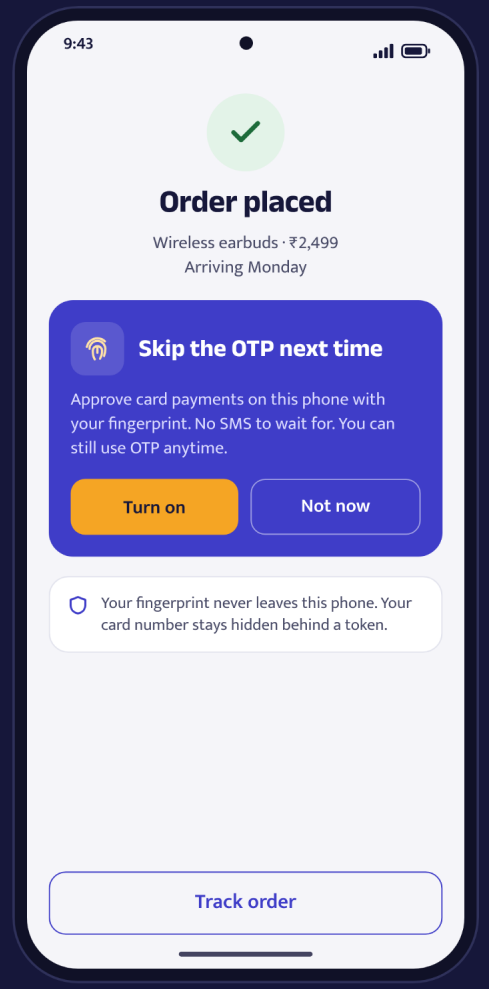
PRODUCT
Mastercard online card payments in India

LENS
UX, business model and strategy

DATE
October 2026

BY
Astha Purohit

Independent analysis. Not affiliated with or endorsed by Mastercard.





AT A GLANCE

The teardown in one slide

A new card payment can take 29 typed characters, two redirects and an SMS wait. UPI takes a 4–6 digit PIN.

WHAT I LOOKED AT

Online card payments with a Mastercard in India: the checkout journey, who controls each step, how money flows, and how UPI and RuPay compete.

WHAT I FOUND

Mastercard writes the rules but owns almost none of the screens. The slowest steps, the bank page and the SMS OTP, belong to banks and gateways.

WHAT I'D DO

Turn every successful OTP payment into a passkey enrolment, standardise in-app authentication, and sell card benefits on big carts.

7 steps
New card, with OTP

4 steps
Saved card, with OTP

3 steps
UPI

2 steps
Saved card + passkey

CONTEXT

Payments in India, by the numbers

24.07 bn

UPI transactions in September 2026, about 802 million a day.

12.4 crore

credit cards outstanding in August 2026, up from 11.2 crore a year earlier.

61%

of credit card spending in August 2026 was online: ₹1.23 lakh crore of ₹2.02 lakh crore.

UPI owns everyday payments. Cards still carry bigger, credit-based and international spending, and most of that money moves through online checkouts.

Sources: NPCI data via Outlook Money (2 Oct 2026); RBI data via Angel One (Sept 2026).

PRODUCT

Mastercard is a network, not a bank

Five parties touch every card payment. Mastercard sets the rules and runs the network, but others own the screens people see.

Shopper

Pays ₹2,499 with a card

OWNS

Their phone, the card they pick and the SMS inbox

→

Merchant

The shop's app or website

OWNS

Checkout screens and saved cards

→

Gateway

Payment gateway and acquiring bank

OWNS

Payment page, routing and SDKs

→

Mastercard

The card network

OWNS

Rules, tokens and the passkey standard

→

Issuing bank

The bank that issued the card

OWNS

OTP page, approval and fraud checks

HOW MONEY MOVES

The merchant pays a fee on each card payment (MDR, typically 1.5–3.5% on credit cards). Most of it goes to the issuing bank; Mastercard earns network fees from the banks.

USERS

Why people still pay by card online

Spread a big purchase	No-cost EMIs on phones, laptops and travel
Earn something back	Reward points, cashback and lounge access
Pay later	Up to about 50 days of interest-free credit
Pay abroad	International websites, apps and subscriptions
Get protection	Disputes and chargebacks if an order goes wrong

UPI'S JOB

95%+

of UPI merchant payments are ₹2,000 or less.

UPI wins small, everyday payments from a bank account. Cards win when the purchase is big, borrowed, rewarded or abroad.

Source: NPCI via Deccan Chronicle, Sept 2026

JOURNEY

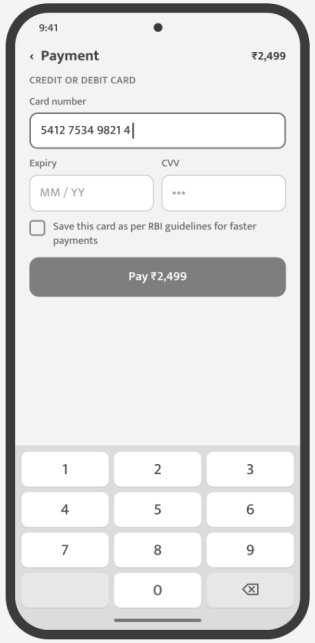
Paying ₹2,499 online: four paths

	New card	Saved card	UPI	Saved card + passkey
Steps for the shopper	**7**	**4**	**3**	**2**
Characters typed	**29**	**6–9**	**4–6**	**0**
Leaves the checkout	Bank page, then back	Bank page, then back	UPI app, then back	**No**
Waits for an SMS	Yes	Yes	No	**No**

Typical Android flow for a domestic card with RBI tokenisation. Saved cards skip the CVV where the merchant supports CVV-free payments; otherwise add 3 characters. Steps vary by merchant, gateway and bank.

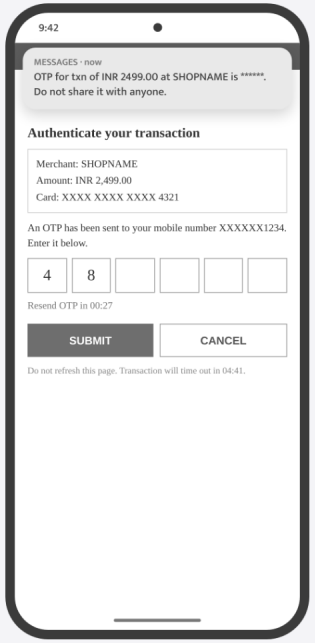
JOURNEY · SIMPLIFIED RECREATION

Card checkout, screen by screen



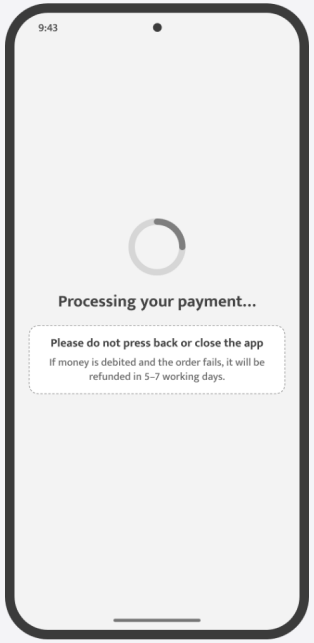
1 · Card details

16 digits, expiry and CVV on a phone keypad, plus a consent line in legal language.



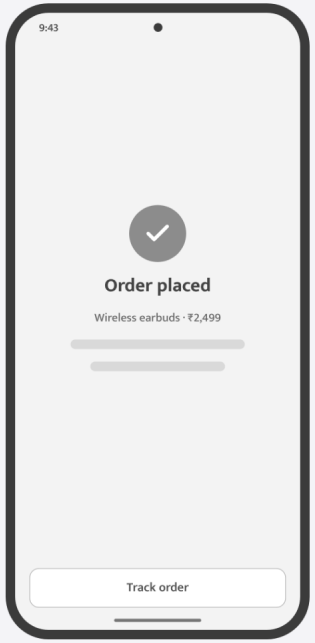
2 · Bank page and OTP

A different-looking page from the bank, then an SMS to wait for and copy.



3 · Processing

‘Don't press back’ and refund warnings make people scared to retry.

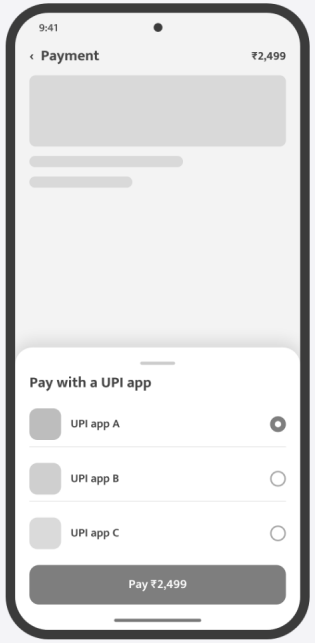


4 · Order placed

Saved cards skip step 1, but the bank page and OTP come back every time.

JOURNEY · SIMPLIFIED RECREATION

UPI, the benchmark



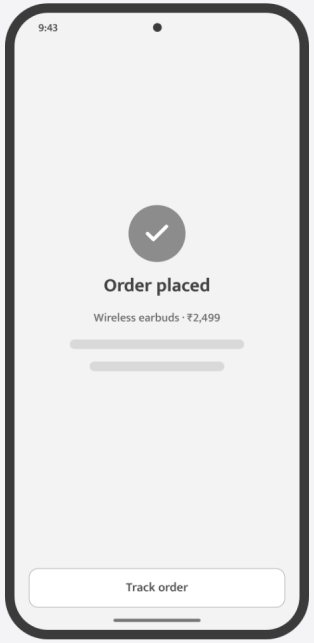
1 · Pick a UPI app

The app opens with the payee and amount already filled in.



2 · Enter UPI PIN

A 4 or 6-digit PIN from memory, the same step in every UPI app.



3 · Order placed

Back to the shop. No SMS, no bank page, nothing to wait for.

WHY IT FEELS FAST

3 steps

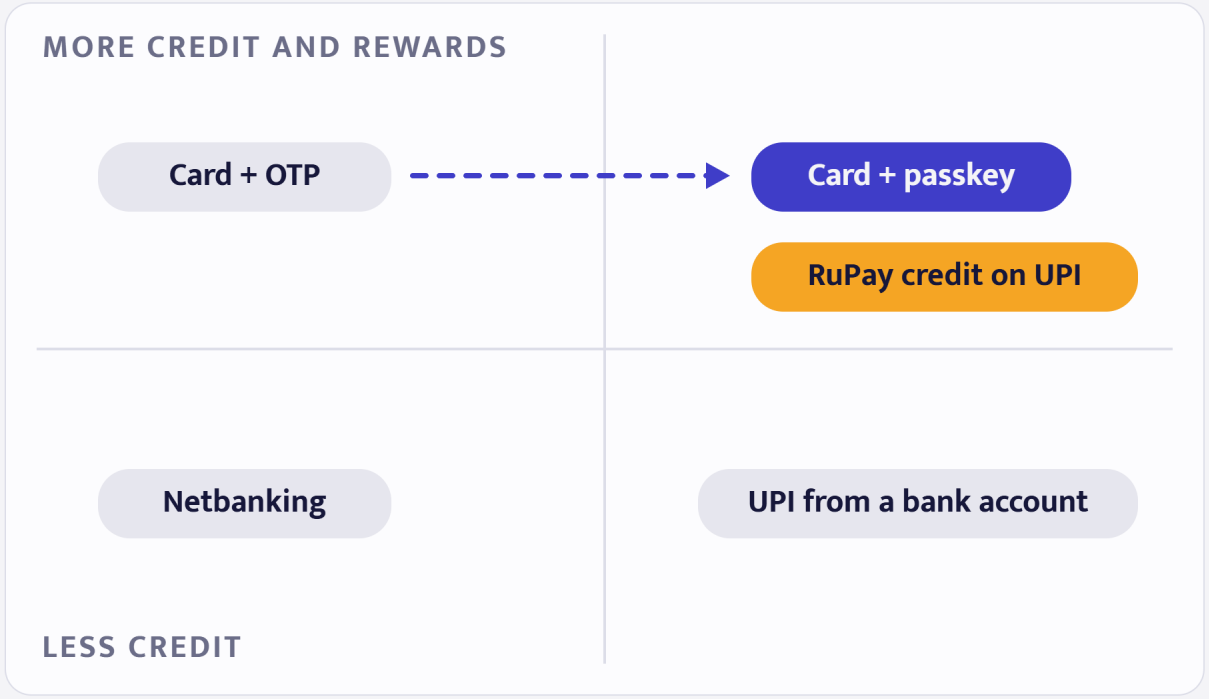
4–6 digits

0 SMS waits

The gap is waiting, not tapping. UPI never asks you to wait for a code.

COMPETITION

Where Mastercard checkout sits



Qualitative positioning: my analysis.

The threat: credit on UPI

RuPay credit cards can be linked to UPI, at 23 banks as of March 2026. Visa and Mastercard credit cards still couldn't be.

The response: passkeys

Passkeys move card checkout to the right without giving up credit, rewards or protection.

The wildcard: UPI access

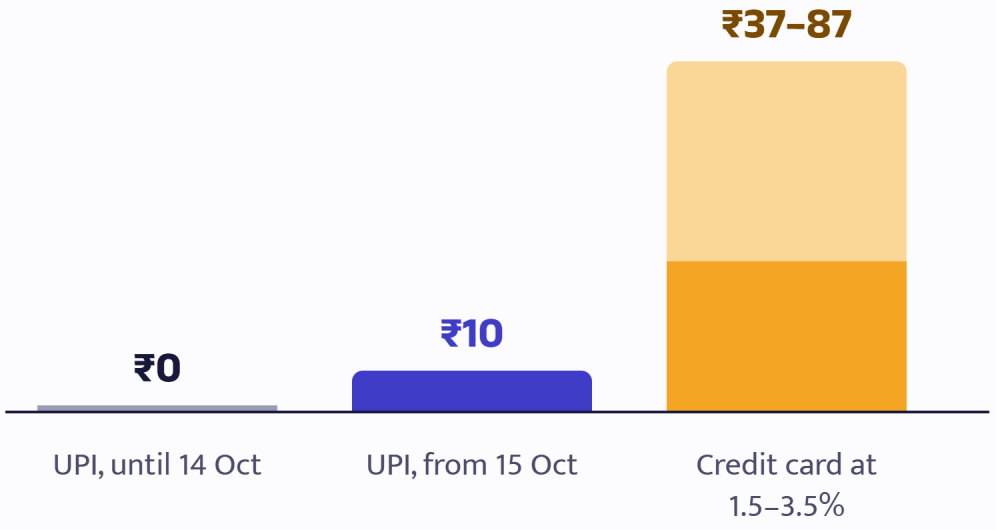
Opening UPI to all card networks had been discussed for over a year without a launch.

Source: CardTrail, 22 March 2026

BUSINESS MODEL

Big UPI payments now carry a fee

Merchant fee on a ₹2,499 payment



From 15 October 2026, merchants pay 0.4% on UPI payments above ₹2,000, capped at ₹300. Smaller payments, over 95% of UPI merchant volume, stay free.

WHAT IT MEANS FOR MASTERCARD

UPI is still far cheaper for merchants, so cards can't win on cost. They have to win on credit, rewards and protection, and stop losing shoppers at the OTP step.

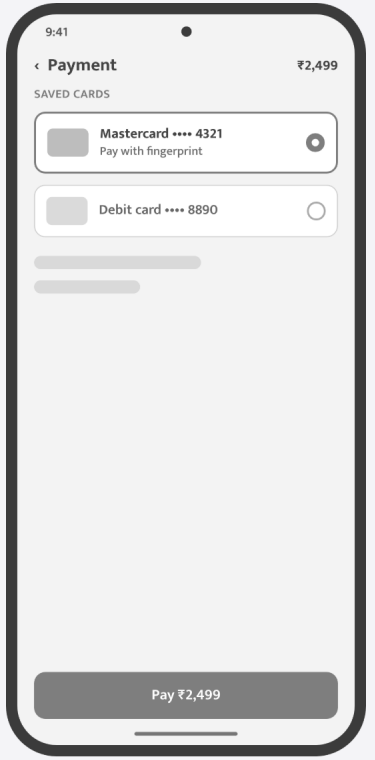
Sources: NPCI framework via Deccan Chronicle (15 Sept 2026); card fee ranges from PhonePe Business (June 2026). Bars to scale.

PRODUCT

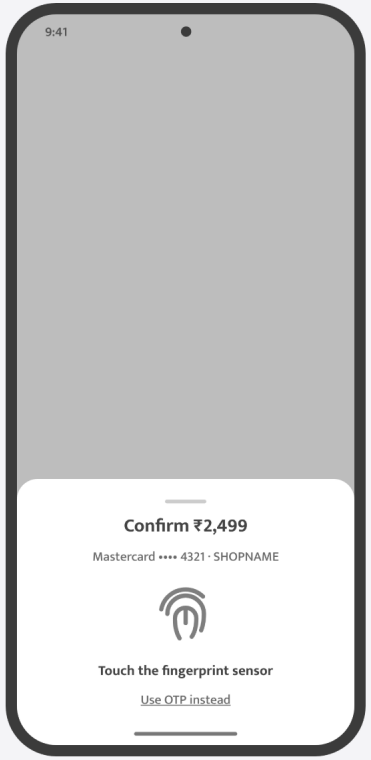
Mastercard's answer: Payment Passkeys

Aug 2023	CVV-free payments for tokenised cards: the CVV is entered once, at the first payment.
Sep 2024	Payment Passkey pilot announced in India, with Juspay, Razorpay, PayU, Bigbasket and Axis Bank.
Mar 2026	Razorpay launches passkey checkout with Mastercard, the first card network to offer it in India.
Apr 2026	RBI's new authentication rules take effect, recognising biometric and device-based factors beyond SMS OTP.

Sources: Inc42 (Aug 2023); Cross-Border Magazine (Sept 2024); Razorpay (31 Mar 2026); RBI directions via The Week (Sept 2025).



1 · Pick a saved card



2 · Touch to pay

PRODUCT

Why OTP still wins by default

WHERE PAYMENTS FAIL

~35%

of payment failures in India start at authentication: late or mistyped OTPs and redirect errors.

RBI still allows SMS OTP, so nothing forces anyone to switch.

Sources: Razorpay (Mar 2026); RBI directions via The Week (Sept 2025)

Three parties must switch it on

The bank, the gateway and the merchant all need passkey support. Any gap falls back to OTP.

First use still needs an OTP

A passkey is usually created after a normal OTP check on that phone.

New phone, new setup

Payment passkeys are tied to the device, so switching phones can mean setting one up again.

Familiar feels safer

OTP feels safe because it's known. ‘Pay with fingerprint’ needs a clear reason and reassurance.

RECOMMENDATIONS

Three bets for Mastercard

BET, IN PRIORITY ORDER	PROBLEM IT SOLVES	IMPACT	EFFORT	BUILT BY
1 · Enrol after a good payment Offer a passkey on the success screen of an OTP payment.	Few people set up passkeys before they need one	High	Low	Mastercard SDK, gateways, merchants
2 · One in-app authentication screen A Mastercard-standard native screen instead of bank redirect pages.	Bank pages look different, load slowly and break trust	High	High	Mastercard with issuing banks
3 · Sell the card on big carts Above ₹2,000, show EMI, rewards and protection next to the card option.	At checkout, cards look like the slow, costly option	Medium	Low	Merchants and gateways

Outside the product: keep pushing for Mastercard credit cards on UPI. It's the one change that would close the gap entirely.

BET 1 · MY CONCEPT SCREENS

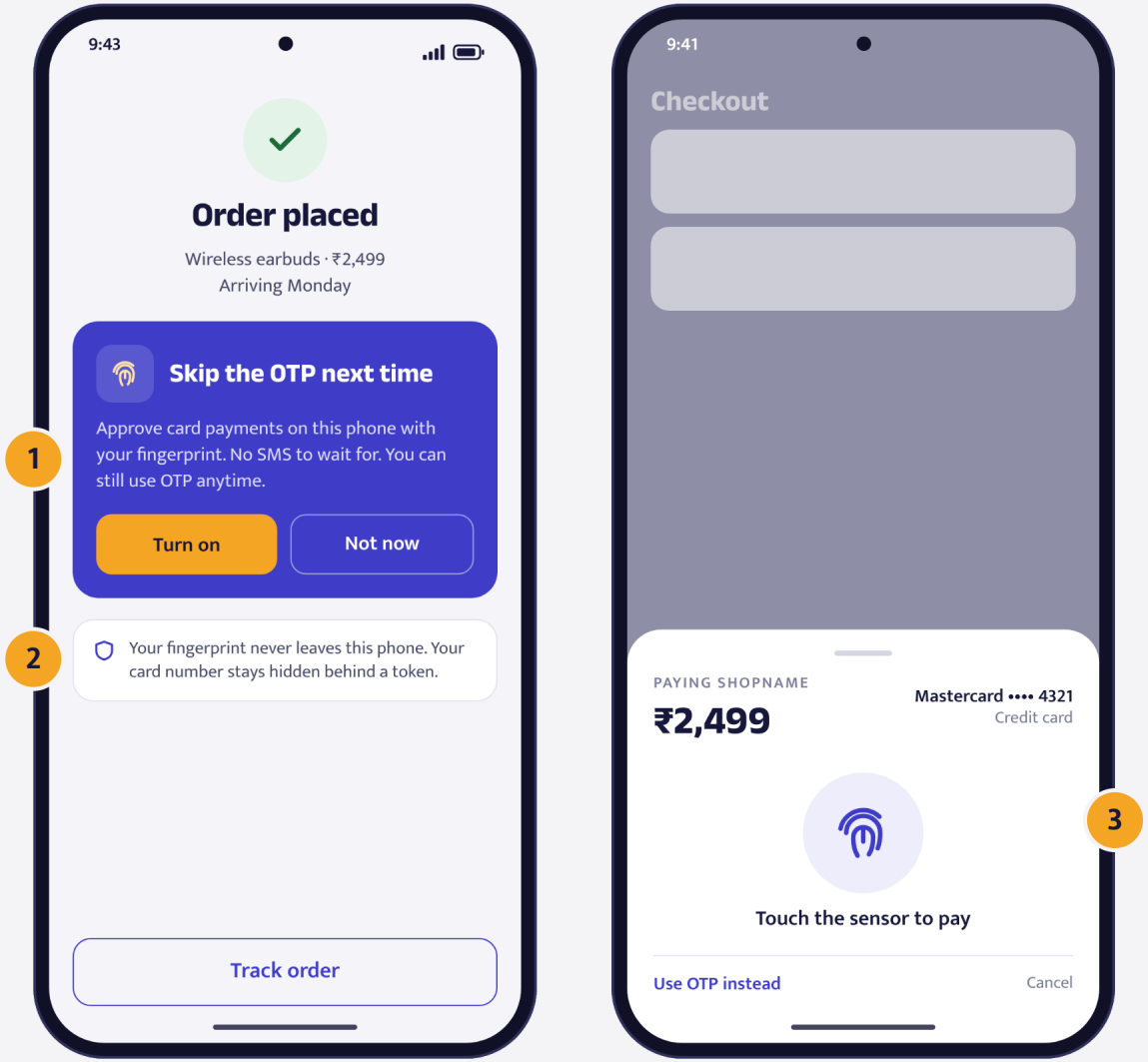
Enrol after success

The best moment to offer a passkey is right after a payment works, when trust is high and the OTP wait is fresh.

1. **Ask after it worked**
   The offer sits on the order-confirmed screen, never in the way of paying.

2. **Answer the safety question**
   One line says the fingerprint stays on the phone and the card number stays hidden.

3. **One touch next time**
   The next payment is a fingerprint, with ‘Use OTP instead’ always visible.



METRICS

How I'd measure it

NORTH STAR

Card checkout success rate: card payments completed ÷ card payments started

ADOPTION

Passkey enrolment rate after a successful OTP payment, and the share of card payments approved by passkey.

SPEED

Drop-off at the authentication step, and median time from tapping ‘Pay’ to confirmation.

GUARDRAILS

Fraud and chargeback rates, false declines, and ‘debited but failed’ complaints.

Test plan: A/B test the enrolment prompt with one gateway's merchants for four weeks. Ship if enrolment rises and fraud doesn't.

TAKEAWAYS

What I learned

Mastercard sets rules, not screens

The slowest steps belong to banks and gateways, so its strongest levers are standards and SDKs.

UPI's edge is no waiting

No SMS, no bank page and one familiar PIN step. And from 15 October 2026, big UPI payments aren't free for merchants.

Passkeys are a rollout problem

The technology works. Adoption depends on asking at the right moment, across banks, gateways and merchants.

IF I WERE THE PM

I'd start with bet 1. It's cheap, needs no new regulation, and every successful OTP payment becomes a chance to move someone to passkeys.

SOURCES

Sources and notes

UPI volumes for September 2026 (NPCI)
Outlook Money, 2 October 2026

Credit card data for August 2026 (RBI)
Angel One, September 2026

UPI merchant fee from 15 October 2026 (NPCI)
Deccan Chronicle, 15 September 2026

Card and UPI fee ranges for merchants
PhonePe Business, June 2026

RBI authentication directions, from April 2026
The Week, 26 September 2025

Mastercard Payment Passkey pilot in India
Cross-Border Magazine, 4 September 2024

Razorpay biometric passkey with Mastercard
Razorpay Newsroom, 31 March 2026

Mastercard CVV-free payments for tokenised cards
Inc42, 3 August 2023

RuPay credit cards on UPI: 23 banks live
CardTrail, 22 March 2026

Notes
Counts come from the typical flow, not timed sessions.

Screens are simplified recreations of typical flows, plus my own concept screens. Independent analysis for my portfolio; not affiliated with or endorsed by Mastercard.

MASTERCARD CHECKOUT TEARDOWN

Thank you

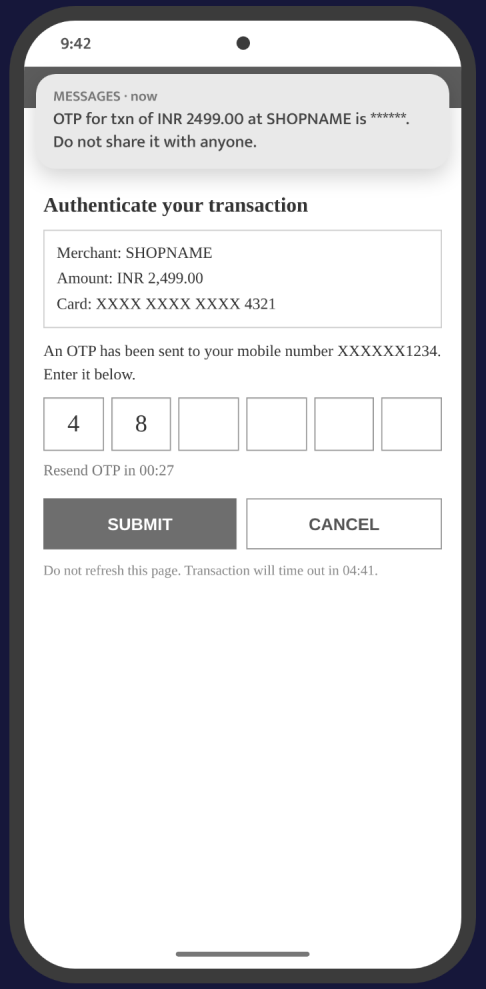
From a bank page and an SMS wait to a single touch. Happy to talk through any of it.

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LinkedIn · Portfolio

TODAY



WITH PASSKEYS

